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COHERENCE AND TRUTH

ALL of this scrutiny of the scientific process got started and has been driven along by the question, Why believe that what science says of the world is true? This is the appropriate time to face up to the question and call on the description that has been built of how science moves to gauge its movement toward the truth. The question is of the accuracy of the scientific description of the world. To put it in a more realistic form and in a form more amenable to constructive answers, ask which aspects of the scientific description are accurate. And, most important as a general evaluative tool, we are interested in knowing a method by which to tell the true from the false. What is it about the process of science that we should attend to as the reliable, responsible process of justification that issues warrant for belief?

This question has been refined somewhat in light of the guidelines that have developed through the description of science in action. From reviewing the practical and conceptual limitations on the activities of explanation, confirmation, and observation, restrictions on the nature of justification have emerged. There is no compromising on the goal, though. Justification is still supposed to be the indicator of truth, or at least likelihood of truth, and there is nothing fancy or philosophical in the notion of truth. It's just plain old truth we are talking about. A theory is true if it is an accurate representation of some part of what there is in the world, that is, of what kinds of things there are and how those things behave on their own and in interactions with us. A theory, like any belief, any proposition, is true if it corresponds to the facts, to the real world. By truth in science we mean exactly the same thing as truth in a court of law. The purpose of a trial and the guidelines of conduct are to find out if the defendant did in fact commit the crime and how. Testimony and verdict are true simply if they accurately report on what happened. Juries and judges don't create the truth by consensus or decree; they must discover it like the rest of us. And so it goes with scientists. The crux, of course, is knowing how to recognize the truth when it comes your way. That's justification.

Though truth *is* correspondence with the facts it cannot be *recognized* by its correspondence. We cannot rely on the facts to guide proofs of scientific theories since the facts are irretrievably at the outer end of the correspondence relation. We could tell that the correspondence link between theory and fact was secure only if we could check both ends, that is, describe and compare both ends. But the fact end, we have learned, is not conceptually accessible in this way, and so relying on the facts is not a useful approach to gathering evidence. The link of correspondence is what we are after but it cannot be directly evaluated, and any useful process of justification must respect this restriction.

So any indicators of the truth of a theory must be internal. That is, they must be within the conceptual and theoretical system of the group of scientists if not of each individual investigator. Justifying factors must be features of which we can be aware. Thus justification must be found in relations among items in the conceptual system, that is, relations among beliefs or, in the more public context of science, relations among theoretical claims. All of the pieces of the puzzle of justification must be recognizable, evaluable, and relevant to the theoretical claims to be justified. The pieces must all have a place in the theoretical system. The process of justifying, then, is a process of comparing aspects of the system, and the accomplishment of justification is the demonstration of coherence among the aspects.

With these guidelines and their genesis in the description of the activities of science, the question of the truth of scientific theories becomes more specific: What does coherence have to do with truth? Why does a coherence among theoretical claims and theory-laden observation claims indicate a correspondence between those claims and the world? Why should links of any kind among theories be a sign of links between the theories and facts? Coherence among theories will secure a cozy network of cooperation and consistent beliefs, but there is no obvious reason to think it will secure any anchor to reality. A requirement of internal coherence, if it is to be a requirement conducive to truth, must be shown to restrict fanciful invention of systems of theories that have no relation to the real world. This is a worry about the possibility of coherent fairy tales, a worry made explicit in 1934 by Moritz Schlick:

> If one is to take coherence seriously as a general criterion of truth, then one must consider arbitrary fairy stories to be as true as a historical report, or as statements in a textbook of chemistry, provided the story is constructed in such a way that no contradiction ever arises. [Quoted by Israel Scheffler in *Science and Subjectivity*, p. 102]

It is a legitimate concern that coherence is too easy, too liberal a standard to be the mark of that unique system of theories that is the true account of the world. It is a worry to take seriously.

THE DIALECTIC BETWEEN THEORY AND OBSERVATION

It is important to consider the activity of justifying scientific theories, the endeavor of increasing the overall coherence of the theoretic system, as an ongoing process. The scientific description of the world is not a static system of claims, and the point of justification is not to show that the current snapshot showing all of today's theories shows nothing but the truth. A model of justification must acknowledge the changes occurring in a theoretical system. Theories come and go and are altered, and most importantly, observational beliefs are steadily added and must be acknowledged and dealt with. The changes, if things are working properly, will be generally in the direction of overall coherence. That is, decisions on what changes to make in the system of theories and on how to account for and accommodate new (and old) observations are made with a mandate to free the system of inconsistencies, to link components and sections of the system by explaining some claims in terms of others, and generally to show that seemingly disparate parts of the picture fit together. The achievement of microbiology, as an example of a recent and significant change in the theoretical system, was to demonstrate a harmony and relevance between the lifeless claims of chemistry and molecular physics and the network of theories and observational claims describing things biological, genetic, and evolutionary. By bridging these domains and establishing the relevance of chemistry to life, microbiology scores high on the virtue of fruitfulness. It promotes coherence in the system by stitching together a great variety of phenomena with a minimum of theoretical thread.

The changes to a theoretical system are not only to promote a neater fit among theories but also to accommodate and influence the accumulating observations. This is the dynamic stimulus to the system and the greater challenge to the maintenance of coherence. The task of science is to deal with the input of observation and revise the system of theories as necessary in the direction of greater coherence. The question to be asking at this stage of the analysis is whether the direction of coherence is the direction of truth. In the ongoing process of justification by coherence, what reason is there to think that science is headed truthward?

The justification process in science clearly pivots on the activity of observation. Even with the burden of theory and conceptual influence, observations are still the link to the environment and will be, if anything is, relief to the worry about coherent fairy tales. Observations must play an active and influential role in blocking any significant warrant for belief in creative, artifactual systems of theories that purport to describe the world but in fact are wildly inaccurate.

This is speaking as if there is a significant difference between observations and theories or between observational claims and theoretical claims, but the work done in Chapter 6 was to show the similarity between theory and observation. This calls for some clarification. Theories and observations are significantly similar in terms of their accountability. Both kinds of claims or, if unspoken, both kinds of beliefs, theoretical and observational, are *of* something, and this of-ness, the descriptive, conceptual meaning of the belief or claim, is based on a relation to other claims, beliefs, and theories. The descriptiveness of a theory or an observation is secured by its fit into the current conceptual network. In this important sense, observations and theories are the same sort of thing.

They are alike as well with respect to their justification. Neither theory nor observation is admissible into the system of science without proper credentials to indicate its accuracy and authority to report what it claims. And, for both theory and observation, that authority is based in a cooperation with other components of the theoretical system. Theories are explained, implied, or sanctioned by other claims in the system. Observations are supported by accounting theories, that is, by other claims in the system. In this other important sense there is no distinction between observation and theory.

In another sense, though, there is an important difference between theories and observation. That is in the genesis of the beliefs, the source of the ideas. Theories and theoretical claims are the result of reflection. Their origin is in thought in that they are produced through inference or suggestion from other ideas, other claims. The idea that the sun maintains its supply of energy from an ongoing process of nuclear fusion at the core is an idea we come by through thinking about things, putting two and two together, so to speak, to see what our other beliefs about energy and astronomy suggest about solar energy. The claim about solar fusion is not caused by any *particular* sensation, though in their lineage many of the claims from which the theory about the sun's energy has been inferred will themselves be linked to sensations. But the claim about the sun's source of energy is one that can be thought of with our eyes closed. It is the pure product of the activity of theorizing, of thinking about, following suggestions and implications of other ideas.

Contrast this with beliefs that enter the conceptual system in a spontaneous way. Many beliefs come to mind not from an active process of inference but as the direct causal result of interaction with the world. Observational beliefs are each one caused by a particular sensation. Their assertive content, that is, the conceptual description that provides a handle on what the sensation is of, and their accountability as reliably caused beliefs are established in relation to other beliefs and theories. The spontaneous beliefs from observation must be dressed in theory to participate as evidence, but they need not be motivated by theory. Observations are forced upon us by the environment. They are the spontaneous effects of particular sensations.

Recall the cosmic rays that present themselves as streaks of vapor in a cloud chamber. A glance at the chamber, crisscrossed with fleeting vapor trails, may give us the idea that there are cosmic rays raining through the room. Describing the observation as being of cosmic rays, associating that descriptive concept with the sensation and sanctioning the view as an accurate source of cosmic-ray information, is a matter of fitting into a web of theory. An appeal to accounting theories describing the causal, informational link between a cosmic ray and the visible track and describing the conditions for viewing the track is what licenses the report as useful evidence. This accountability indicates that what we come to believe by looking at the cloud chamber we believe truly. But we come to believe by looking. We got the idea about cosmic rays in the room in the first place as a result of the particular sensation. The belief is caused by an influence from the world, not by an active process of thought. It was a spontaneous belief, an observation.

This marks an operative difference between observational claims and theoretical claims, between observation and theory. It is a difference of degree since many scientific claims are the product of reflection about particular sensation and so are hybrids between observation and theory. There is nonetheless this useful distinction based on the cause of a belief rather than its descriptive content or source of accountability. Observations are the result of causal, sensory input from the environment. The objects and events in the world do not directly justify the scientific claims we make, but they do cause many beliefs, those that are the spontaneous results of particular sensations. This is the distinction and the importance of observation.

The evidential authority of an observation is based, as has been said, in its accounting theories. Given the pivotal importance of observation as evidence, it is clear that the justification process wants the best accounting theories it can get. Much of the weight of the question of believability of science rests on the accounting theories. Why believe *them?* In fact, the work of analyzing justification has only been postponed to this point and focused on this link between observational evidence and the theories for which they stand. How are the accounting theories confirmed? Like any other theory.

Accounting theories are nothing special. They are distinguished by their manner of use, not by their content or degree of generality and not by their manner of confirmation. Any theory can be used in the role of accounting. In one context a theory may be the valued end, the accomplishment of description. Theoretical claims in optics, for example, are simply nice to know. Light and its behavior are part of the world, and so optics is part of the accomplishment of science. But the theories of optics are also nice to use. They function well as accounting theories that keep track of the informational flow from the environment to ourselves. The same is true of theories of neutrino physics. They are useful as accounting theories, describing observations of what is happening in the solar interior, and they are in their own right informative theoretical descriptions of an aspect of the world. Their being *accounting* theories is a circumstance of their use on a particular occasion. So asking how an accounting theory is confirmed, asking how we know what we do about neutrinos, for example, is exactly a return to the general question of confirmation.

Accounting theories, like all theories, are confirmed by their plausibility relative to other theoretical claims and by their coherence with observational claims. Theories of optics or neutrino physics, active players as accounting theories, are confirmed in the familiar pattern of explaining phenomena and making successful predictions. Theorizing about light waves and their interactions explains all sorts of common phenomena from the colors of the rainbow to your short, fat legs as you stand waist-deep in a swimming pool. Theorizing about neutrinos explains what happens to the energy in some otherwise mysterious interactions between elementary particles. These theories, employed here and there in support of observations, have their own explanatory and predictive successes. They too confront the observational data and thereby establish their justification.

The general picture that has emerged of justification in science is one of mutual support between observations and theories. Theories, whether or not they find a use as accountants, are understood and justified by reference to observations. But then, observations are understood and justified by reference to theories, accounting theories in particular. That is, observations are accountable to theories. Neither sort of claim is the single, basic source of justification in science. It is rather that the justification derives from a mutual support between the two. This may sound vaguely circular, and it is vaguely circular. But it is not a problematic circle in the sense of immediately impeaching the process or the evidence. To see why the circle is tolerable (as well as inevitable) it is better to think of the relation between theory and observation as one of negotiation.

Theory and observation, in the activities of explanation and confirmation, engage in mutual justification. It is a process of give-and-take between the two, of adjustments to either side, but always under the constraint to increase coherence. Scientific theories report the big picture of the world. They provide a global-scale understanding of what sorts of things there are in the world and how they interact. This understanding then allows us to make sense of sensations. Observations, given meaning and credibility by theory, report on individual situations in the world and provide a more local-scale description. The observations then constrain theorizing since theories must be compatible with observations. And more, theories must explain and show cause for observations. Even with the complication and ambiguities inherent in explaining and confirming, with the necessity of auxiliary theories and questionable conditions of testing, in all cases of comparing theory to observation the result must be compatibility. Among all the participating claims, the theory being tested, the auxiliary theories, the observation claims, justification requires a thorough compatibility. If a prediction or explanation fails, that is, if theoretical claims are in conflict with observation, *something* has to give. The goal is coherence, and it begins by purging inconsistency.

The influence between theory and observation goes back and forth. It is not always in the same direction, say, observation forcing the ways of theory. With a steady input of new observations the system of scientific claims must be kept open for revision to insure a fit for new observation claims. These new observational beliefs cannot be ignored. If they cause trouble by contradicting or offending or casting doubt on theories in the network, then a readjustment of the network or of the observation itself is in order. You do what it takes to move the developing system of claims in the direction of greater coherence. You do what it takes to tighten the explanatory links between claims. And you do anything it takes to free the system of inconsistency.

This is the most general view of the process of scientific justification. It is an ongoing process of accommodating new observations within the system of theories and doing so under the constraint of enhancing coherence. It is surely not a monotonic rise in coherence that takes place in real science. We may pay with incoherence in one part of the system for a tighter fit somewhere else. Or we may provisionally accept new ideas that bring disarray, on the hunch that they will eventually precipitate a greater coalescence in the system. No one expects each step to be an enhancement of coherence, but there is good reason to expect a long march to be generally in this direction.

A consequence of this processual view of science is the expectation that individual theories and whole patches of related theories can change over time to allow an increased coherence in the larger system. These changes could result from the acceptance of new observations that threaten old beliefs. Changes in the theoretical system could also be the result of clever thinking about old ideas, discovering ways to find agreement between disparate patches within the system. That is, it is not only observations that can cause a reshuffling of theory; the recognition of imperfections within the system of theories can do it as well. Our current network of theoretical claims about the world is not perfectly coherent. It is modular in the sense of having several seemingly unrelated patches, and it is likely always to be this way. There is a nest of claims about relativity and gravity, for example, that has nothing to do with the group of claims about the evolution of life. There are even collections of theoretical claims within the system that appear to be inconsistent. Quantum mechanics and the theory of relativity, two of the best and brightest in physics today, do not always get along. As described by general relativity, the material at a spacetime singularity (as described in Chapter 1) must be smashed to a single point, infinitely concentrated and precisely located. A quantum-mechanical description of the physical world explicitly prohibits any such point locality of material stuff. So what are you going to do? Even with no new observations to consider it is clear that the theoretical system wants adjusting for consistency and explanatory closeness, which is the achievement of coherence.

Einstein's development of special relativity is a nice case of changes to the theoretical system with the motive and result of reconciling differences between theories and enhancing the general coherence. The central idea of relativity, that no frame of reference is distinguished as being stationary in the universe, that all frames, in other words, are created equal, was certainly not new with Einstein. What was new was the adjustment to the theoretical system to smooth the inconsistency between this principle of relativity and the new but phenomenally successful theory of electromagnetism. The principle of relativity insists that no possible measurement can be performed that would distinguish one uniformly moving reference system from another. The equations of physics are exactly the same on a moving train as on the stationary ground. The theoretical account of electricity and magnetism, though, seemed to suggest that the effect produced by moving past a stationary magnet would be different from that produced by moving the magnet past a stationary observer. The electromagnetic theory, in other words, distinguished a preferred frame of reference, namely, the one in which the magnet is stationary.

The accomplishment of special relativity is to reconcile this difference by changing the notions of time and space, the media of measurement, to fit the theory of electromagnetism into the framework of relativity. By saying that the measurements of space, time, and mass are dependent on the relative motion of the measurer, Einstein sacrificed entrenchment to resolve an inherent inconsistency and to enhance the coherence of the theoretical system.

Einstein's revisions to the understanding of measurements of space and time also allowed the theoretical system to accommodate the new observations of the Michelson–Morley experiments. Michelson and Morley, at about the same time that Einstein was thinking up the special theory of relativity, devised and performed an experiment to detect the effects of the earth in its travels around the sun, moving through the special rest frame as suggested by the electromagnetic theory. They found no effect. The experiment was repeated under various conditions, and in various locations to the extent that its null result could not be blamed on experimental conditions. The observation, or the expected observation, was certainly heavily theory-laden. It's not that a sufficiently sensitive observer feels a breeze as the earth plows through the sea of ether that rests in the electromagnetic frame, nor can we see any movement relative to stationary points of reference. To detect the motion one splits a beam of light into two and sends the two beams at right angles and notes if one has taken a slightly longer time to travel an equal distance, longer because it was slowed by fighting the current of the ether. The comparison to travel times is done by reflecting and recombining the two beams and noting the bands of light and darkness that are caused by the wave properties of the light and are influenced by the relative arrival times of the two waves. It is not a simple experiment, and the observation that there is no motion of the earth relative to the ether depends heavily on accounting theories to link the final fringes of light (the things we observe) to information about the motion of the earth (the event we are interested in). In fact, attempts were made to accommodate the refuting evidence by blaming the accounting theory. The effect (the earth's motion through the ether) is really there but we are just not seeing it with this Michelson–Morley apparatus because the pressure of the ether on the leading edge of the table holding the optical equipment actually compresses the table so that the light that must fight the current has a shorter path to travel. The shorter path at slower speed just compensates, and that is why the two beams of light arrive at the point of detection at the same time.

This bit of "ad hocery" never really caught on. But in any case, it is clear that the Michelson–Morley observations could not be ignored. They forced a negotiated agreement among the observation report, the accounting theories, and the theory being tested. Einstein's theory of relativity suggested a change in that part of the electromagnetic theory that mentioned or required a preferred reference frame or a medium such as ether to support the electromagnetic effects. It modified as well scientific beliefs about measurement in space and time. The net effect was that a null result of the Michelson–Morley experiment was to be expected. It's not that Einstein's theorizing was motivated by or in response to the Michelson–Morley experiment. Rather, both the theory and the observation gain credibility by their compatibility and by the newly demonstrated harmony between the theoretical description of electromagnetism and the principle of relativity.

The point is that an applicable model of the process of justification in science must have room for this aspect of change in the theoretical system. The change, as in the case of special relativity, is swept by the tide of coherence. In the case of special relativity, basic consistency among theories is gained by the introduction of novel and rather startling (a polite way of saying prima facie implausible) ideas of the nature of distance and time. As the network of theories changes and as new observations are accommodated, justification changes. The special theory of relativity replaces the Newtonian description of motion, but that does not mean there was never justification for Newtonian mechanics. Nor does it mean we will always be justified in the special relativistic mechanics. Under the circumstances that preceded the

Michelson–Morley experiments and the clear development of the description of electromagnetic radiation, the Newtonian view was justified. But under current circumstances, that is, in light of the current network of theories, the relativistic view is justified. Scientific justification must be contextual justification, sensitive to the circumstances of the time. The alternative is to put off all evaluation of justification until the end of science, whatever that means, and see what stage in the development showed the greatest coherence. But that is doing nothing at all.

The process of science presents a steadily developing picture of the world, a picture whose coherence is maintained and even enhanced by an ongoing negotiation between theory and observation. Stages of the process are often begun by speculation, taking a guess about what the world is like, to break into the cycle of theory in conceptual support of observation and observation in evidential support of theories. Hypotheses, the currency in the context of discovery, are needed to get started since a theoretical framework of some kind is necessary to do anything worthwhile, to think, to observe, to test. But the hypotheses used to break into the cycle are not irresponsible, carefree speculation. They are accountable. They must be tested for coherence with other theories and with observations. They and other theories are subject to revision and are used to account for observational evidence, which in turn directs revisions in the system and perhaps the suggestion of new theories that will influence and be influenced by observations.

READING THE BOOK

This process of negotiation between theory and observation and the account of the developing theoretical picture of the world bring out in full the methodological similarity between science and reading a text in an unfamiliar language. An unfamiliar text can be deciphered by a process often referred to as the hermeneutic circle. It is **hermeneutic** in that its goal is interpretation of the text, and it is circular in open acknowledgment of the necessity of using only our available information to account for and extend our available information. It is circular because of its limited resources. There is only the text, the marks on the pages. We have no access to the actual events being described, assuming it is a historical narrative of events long past. Nor can we interview the author and get the straightforward explanation of the plot. Whatever we make of the text must come from what is on the pages and from our own theorizing. Initially, of course, the marks on the page are meaningless to us, and the larger message of the text is unknown.

The hermeneutic method of interpretation is very similar to the scientific method of understanding the world. By describing this similarity I do not mean to suggest that it is anything more than a methodological similarity. That is, there is no suggestion that the objects of natural science are meaningful or symbolic as are the linguistic marks of textual exegesis. Nor is there any hint that nature must have an author as does a text. Understanding a book involves understanding the intentions of the author, and intentions are particularly illusive. The meanings of words, as we know from Chapter I and from our own experiences, are under the control of people, including authors and those who influence them. There is no tight connection between the words printed on the page and what the author intends them to mean. It is a connection that is influenced by the cultural and intellectual environment of the author, and a connection that is not immune to the vagaries of the writer's caprice and whimsy. These complications must be dealt with in translating texts, as they are under the control of people with who knows what on their minds. These complications are missing in natural science. Nature has no author and does not have to deal with intentions behind the events it understands. The objects of study by natural science and by textual interpretation are very different, but the method of study, as we will see, is significantly similar.

Consider first the hermeneutic method of interpreting a text; then we will see how the method is similar to the process of science. To translate the text one must initially speculate on the meaning of individual words and passages, speculation based on patterns of recurrence of symbols and on their place in the context with other patterns and passages. These initial speculations on the meanings of the words provide preliminary hypotheses about the messages of individual passages. They also lead to a way to test the speculation. We can try out the assumed assignment of meanings on other occurrences of the symbols, other passages in the text, and see if the resulting translation makes sense. If it does, the standards of testing can be tightened by asking whether the newly translated passage is consistent with others that have been tentatively translated already. Stricter still, do the passages as given meaning under the speculated translation hang together? Do they show a continuity and relevance to each other that one expects of a coherent text? If not, if translations of new passages are just gibberish, or they contradict other passages, or the whole thing is turning into just a bagful of unrelated, irrelevant claims, then revision is called for in the initial guess at the assignment of meanings to words, or in the speculative theory about the message of the plot.

The process of translation advances by a back-and-forth exchange of information between the developing understanding of the plot and the translation of individual passages. The global understanding, the message of the whole work, guides the local understanding of the parts, the individual passages. These passages are translated, they acquire their meaning, by their fitting into the bigger picture and making sense. The meaning of the parts is discovered from their context. The understanding of the parts in turn guides the understanding of the whole in that the message of the plot is built from the component messages of the individual passages.

This mutual guidance between the local reading of passages and the more global understanding of the text results in an ongoing negotiation between the two. The reading of the text is constrained by a requirement of coherence. It must, in the end, make sense. The passages must be consistent and should hold together in a cogent message, at least in sizable sections of the text. As the reading continues, new passages are encountered and must be accommodated within the network of beliefs about the book and its message. Each new passage is like a new observation, of which the reader must make sense and which must be fit coherently within the theoretical system.

It is the coherence of the interpretation that makes it believable. If the developing understanding of the message of the book is kept consistent and coherent, there is reason to think that it is developing toward the correct reading. Consistency is certainly a necessary feature of an acceptable, that is, likely-to-be-true interpretation of the text. The standard of consistency then can be used to rule out unacceptable readings. It is a way to falsify some suggestions of the meaning of the text as not-possibly-accurate. But consistency alone is far from sufficient. There will be plenty of consistent readings of the book, alternative assignments of meanings of the symbols that produce alternative messages from the text. Sufficient cleverness and diligence on the part of the reader could produce any number of consistent readings of the text, but only one of them will be the correct reading that conveys the intended message (still talking about a book, not nature). To trim this list of potential readings we must apply additional criteria of acceptability, criteria that can be judged on the basis of information found within the text itself and within our own growing understanding of it. The next steps to justify a proposed reading cannot involve consultation with the author or a glance at the events the book is about. These facts are inaccessible, and we must deal with the information as filtered through the text and as organized by our understanding of the text. That is all we have to go on.

Why is consistency a requirement of acceptability of a theory about what the text says? Why aren't inconsistent readings considered? The answer is based on an assumption that the text makes sense. It is a principle of charity, giving the author the benefit of the doubt on the issue of reasonableness. The book, it is assumed, has been written to be informative. It simply does not deny on one line what it asserts on another. The message is free of contradiction, and so our reading of it must be as well. This is an assumption about the way any book *must* be written. It is not something that can be proven through reading, that is, through observations of texts. It is part of the conceptual preparation we must do to facilitate reading. But like every other claim we make about the world or about the text, this one is bendable. As a last resort, if no consistent reading is developing after diligent hermeneutic work on a text, it may be that contradictions have been planted in the message itself. The inconsistency might be just the way things are in the plot. This option has perhaps occurred to many people trying to read *Finnegans Wake*, as good an example as any of a book written in an unfamiliar language. This has to be, though, the very last resort.

Other standards of justification of proposed readings, that is, other criteria by which to cut the field of consistent readings, are generated by extending the principle of charity. The plot of the book was written not only to be free of contradiction but also to hold together in such a way that many of the passages will be relevant to each other. The plot, we assume, makes some sense in that one thing leads to another and what happens here may explain what happens there. The message, it is assumed, has this feature of organization, at least in sizable sections or chapters, and any accurate reading of the text must reflect this through a coherence, a fitting together of the interpreted passages. This criterion of coherence in the reading of the text is thoroughly evaluable with the information on the page and in our theories of meaning. It is an internal feature of these theories, and so it is a useful criterion to further filter the likely-to-be-accurate readings from the less likely.

In the context of science, where the goal is not to read a text or to comprehend the intentions of an author but to describe nature, a similar principle of charity is appropriate. The dialectic of observation and theory proceeds under an assumption that nature cannot tolerate contradiction. Objects in the world cannot both have and not have particular properties. They cannot both be and not be particular ways. Things are one way or another. This is a requirement of how any world must be, and so overall consistency is a necessary condition for acceptability of a theoretical system.

Nature is more than just free of contradiction; it is organized. Events in the world do not just happen, one damn thing after another. Objects interact in consistent ways, and events at one time and place will be relevant to events at another. What happens here and now may influence what happens there and then. This organization in the world must be reflected in our theories of the world and it is, as coherence. Coherence in the theoretical system serves to filter the likely-to-be-accurate theories from the less likely. In this sense the feature of coherence is truth-conducive. As in the case of reading the text, it may not be warranted to assume that the world is wholly organized. There may well be a disunity of things such that organization and relevance are to be found only in pockets, in domains of objects and events. Our theoretical system then may show a modularity of coherent networks but no global cohesion.

Consistency and coherence in the theoretical system are necessary features of an accurate description of the world. But still these criteria are not strict enough to identify a single theoretical description as the true one. Back in the context of reading a text, there will always be multiple readings and multiple assignments of meanings to symbols that are equally consistent and at the moment equally coherent. Coherence and consistency together are not sufficient to rule out all but the correct reading or, in the case of science, the true account of the world. But this is old news. From the beginning we have been warned that there is no hope of certainty, whether it is in evaluating the correct reading of a text or in justifying scientific theories.

The responsible thing to do in this situation is, first of all, to continue reading. In the context of science this means to continue observing. As more passages must be fit into the plot or more observations must be accommodated, the theoretical system will be forced to adapt to an increasingly higher degree of difficulty of achieving coherence. This will serve to further weed out misfit claims and previously, though no longer, coherent theoretical systems.

The other responsible thing to do is to devise other internal standards by which to judge theories or readings. For example, a false reading of the book or a false system of theories of nature would most easily maintain its consistency and coherence if each particular theory was tested by observations that were interpreted by that theory itself. The tight, insular circularity of acting as the accounting theory of your own evidence would severely limit a theory's exposure to possibly refuting information. Realizing this we can count it as virtuous of a theoretical network if the observations used to test a particular theory depend only on accounting theories that are independent of that theory being tested. Even though all evidence must be accountable by theory, that is, it is all theory-laden, we can insist on *independently* accountable evidence. The observations must be laden with theories that are independent of the theory for which they serve as evidence. This restriction blocks the collusion and close, self-serving circularity of a theory prescribing its own evidence. The standard of independence is a significant addition to the list of internal virtues of theories. It is significant enough, in fact, to warrant its own section.

INDEPENDENCE

Independence is another virtue along with explanatory coherence, entrenchment, and the others. Paying attention to the independence of the evidence used in science means we must always get beyond the slogan that observation is theory-laden and in each case attend to *which* theories are used to influence the observations. It is important to the evaluation of justification to identify the accounting theories in each case and see if *they* have a stake in the outcome of the observation. Here is an example: Theories of optics are used to account for the image formed by an optical microscope. Knowing how light behaves in its interactions with the specimen and the lenses tells us how the image is formed and that the image reliably displays features of the specimen. If the image is of a chromosome, that image may function as evidence of some claim about relevance of shape to function of the object or about relative numbers and sizes of chromosomes in different organisms. The claims about optics that are used to describe the formation of the image and to thereby sanction it as an accurate representation of the specimen are independent of the claims about biology and genetics for which the particular observation functions as evidence. The optical theories do not gain or lose credibility on the basis of the outcome of biological experiments. The accounting theories in this case are clearly independent of the theories being tested by the observations.

Why is independence good? That is, why is this feature of independence between accounting theories and the theory being tested a truth-conducive feature? The goal of validation of scientific knowledge is the elimination of theoretical artifacts. We want the entities and events described by a natural science to be features of nature and not simply creations of theory without correlates in the world. The chances that several unaffiliated theories will cooperate to manufacture artifactual evidence are less than that a single theory will do just that.

An analogy can help motivate this point. Newspapers make descriptive claims about the world, as does science. If I read in the paper a story about events that are inaccessible to me, events in China, say, what makes the story believable? In part, a story is credible if it appears in other media such as independently owned and managed newspapers, radio, and television. We are dubious of a story that is reported by only one newspaper and, to borrow from Wittgenstein, there is little to be gained in terms of objective confirmation by buying and reading another copy of the same newspaper. The point is that the chances of a single newspaper reporting falsehoods, by design or by mistake, are greater than the chances of several independent media reporting the same untruths. The quality of the evidence for a story (or a scientific theory) is measured in part by its independence.

The benefits of independence can be further appreciated by considering our own human perceptual systems. We consider our different senses to be independent to some degree when we use one of them to check another. If I am uncertain whether what I see is a hallucination or real fire, it is a less convincing test simply to look again than it is to hold out my hand and feel the heat. The independent account is the more reliable because it is less likely that a systematic error will infect both systems than that one system will be flawed.

Another analogy to the concept of independence is found in history. Ancient history in particular is fortunate to have available two independent sources of information, the literature of ancient texts and archaeological artifacts. Interpretation of artifacts, their age, their use in trade, and the like, is credible insofar as it is consistent with interpretations of ancient texts. In this way, claims made by archaeologists can receive an independent corroboration by the claims made by interpreters of ancient texts.

There are plenty of good examples in the physical sciences of independence functioning as a rational measure of belief in scientific claims. One of the best is in the experiments of Jean Perrin in the beginning of this century, experiments previously mentioned in Chapter 5, designed to measure the number of molecules in one mole of material, Avogadro's number. (A thorough account of Perrin's accomplishment is to be found in Nye, *Molecular Reality.*) Perrin measured the same physical quantity in a variety of different ways, thereby invoking a variety of different auxiliary theories. And the reason that Perrin's results are so believable and that they provide good reason to believe in the actual existence of molecules is that he used a variety of independent theories and techniques and got them to agree on the answer. The chances of these independent theories all independently manufacturing the same fictitious result are small enough to be rationally discounted. It is the independence that supports the credibility of the account.

The benefits to justification in these cases come from the features of independence and coherence. They are not inherited from any particular theory that contributes to the process. Using theories of optics to account for use of a microscope to observe chromosomes is an example of good, credible evidence, not because the theories of optics are well confirmed but because they are independent from the biological claims being tested. The same is true in the case of using carbon-14 or dendrochronology techniques to determine the age of archaeological artifacts. These techniques supply credible evidence, but not because good, hard sciences are allowing their authority to trickle down to more supple science. More importantly, the evidence is credible because its gathering is independent of that for which it serves as evidence.

With the requirement of independence we can ease what may have appeared to be a tension between two key components in the description of science, the facts that observations are influenced by theory and that these observations are the evidence used to test theories. This relationship need not be circular as the theory that does the influencing need not be the same as the one that takes the test. This circle-blocking independence is a measure of objectivity of the evidence and of the process of justification. Independent evidence is outside of the influence of the *particular* theory it serves, though it is still within the theoretical system of science. It is internal, as it must be, in the latter sense, though external of the particular claims it tests. Furthermore, the evaluation of independence is not influenced by psychological, social, or aesthetic factors. It is objectively measured, out of the control of whimsy and bias.

Independent evidence is objective evidence, and the requirement of independence is a key ingredient of the scientific process that prevents problematic circularity in the justification of theories.